

The BELEBELE Benchmark: a Parallel Reading Comprehension Dataset in 122 Language Variants

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Abstract

We present BELEBELE, a multiple-choice machine reading comprehension (MRC) dataset spanning 122 language variants. Significantly expanding the language coverage of natural language understanding (NLU) benchmarks, this dataset enables the evaluation of text models in high-, medium-, and low-resource languages. Each question is based on a short passage from the FLORES-200 dataset and has four multiple-choice answers. The questions were carefully curated to discriminate between models with different levels of general language comprehension. The English dataset on its own proves difficult enough to challenge state-of-the-art language models. Being fully parallel, this dataset enables direct comparison of model performance across all languages. We use this dataset to evaluate the capabilities of multilingual masked language models (MLMs) and large language models (LLMs). We present extensive results and findings, notably that despite significant cross-lingual transfer in English-centric LLMs, much smaller MLMs pretrained on balanced multilingual data still understand far more languages. Overall, BELEBELE opens up new avenues for evaluating and analyzing the multilingual capabilities of NLP systems.

1 Introduction

The absence of high-quality, parallel evaluation benchmarks is a major obstacle in assessing the text comprehension capabilities of multilingual models. NLP datasets with high language coverage do exist, such as FLORES-200 (NLLB et al., 2022), but they primarily focus on machine translation. Popular multilingual evaluation benchmarks, such as multilingual question answering (Lewis et al., 2020; Clark et al., 2020), natural language inference (NLI) (Conneau et al., 2018), and summarization (Ladhak et al., 2020; Hasan et al., 2021), altogether only cover around 30 languages. And while understanding and generative text services are used across the globe in 100+ languages, the

lack of labeled data provides a major obstacle to building functional systems in most languages.

Simultaneously, large language models (LLMs) have become increasingly popular. Certain LLMs, like BLOOM (Scao et al., 2022), are trained on multilingual data and tout their innate multilingual capabilities. Others like GPT-3 (Brown et al., 2020) and LLAMA (Touvron et al., 2023a) have demonstrated multilingual competence despite their training data being predominantly in English. Even so, LLMs benefit from pretraining data that is linguistically diverse, intentionally or not, as well as from cross-lingual transfer (Zoph et al., 2016; Artetxe et al., 2020; Muller et al., 2021b). But how multilingual are these models really? Beyond LLMs, significant scientific progress needs to be made before NLP systems can be built effectively and efficiently in low-resource languages. Many modeling techniques are being presented as language-agnostic but have only truly been evaluated in a small number of languages (Bender, 2011), risking not being applicable to diverse typologically phenomena (Bender, 2009). We believe that large-scale, parallel, and discriminative datasets are crucial for studying the multilingual capabilities of such models and understanding how the technological disparity between high- and low-resource languages is evolving.

In this paper, we present a fundamental natural language understanding benchmark to evaluate language models across 122 language variants from around the world¹, called BELEBELE². The dataset contains 900 unique multiple-choice reading comprehension passages and questions. The questions have been carefully crafted to discriminate between models with varying competence in language comprehension. While the questions do not necessarily require higher levels of knowledge or reasoning, they favor generalizable NLU models and deliber-

¹Download at github.com/facebookresearch/belebele

²Bambara word meaning "big, large, fat, great".

Language	Passage	Multiple-choice questions			
English eng_Latn	Insects were the first animals to take to the air. Their ability to fly helped them evade enemies more easily and find food and mates more efficiently. Most insects have the advantage of being able to fold their wings back along the body. This gives them a wider range of small places to hide from predators. Today, the only insects that cannot fold back their wings are dragon flies and mayflies.	Q1. An insect's ability to fold their wings back increases which of the following?			
		1. Food supply	2. Hiding spaces	3. Finding mates	4. Flight speed
Tigrinya tir_Ethi	ሓሽራ ናብ ኣየር ዝወሰዱ ኣፈናቂ ኣለዎም ነይሩ። ናይ ምንፍር ከኣላቶም ጸላእቲ ብቐሊል ከጥቅዑን ከምኡውን ምግብን ሙሉዞ ብብቕዕት ክረኽቡ ሓጺዮም። ሙብዛሕኡም ሓላጽ ሙንግብብን ናብ ኣካላቶም ናይ ምዕጻፍ ክእለት/ብልጫ ኣለዎም። እዚ ካብ ሃይጋቲ ክሕብሉሎም ዝኽእሉሎም ብዘሓት ኣረገገተይ ቦታታት ዝህበሎም እዩ። ኣብዚ ሕጺጺ እዮን፣ እቶም ሙንግብብን ከፃጽፉ ዘይከኣሉ እንኮ ሓላጽ ድጋንፍላይብን ሜዕራይብን እዮም።	Q1. ከእለት ናይ ሓላጽ ሙንግብብን ጋድሕራት ከፃጽፉ ምንእል ካብዞም ዝሰዕቡ ነዮንት ይወሰኑ?			
		1. ቐረብ ሙንግብ	2. ሙሕቢኢ ቦታታት	3. ምርኻብ ሙጽምጺ	4. ናይ ምህዳም ፍጥነት
Khmer khm_Khmr	សត្វល្អិត គឺជាសត្វដំបូងគេដែលហោះហើរនៅក្នុងខ្យល់។ សត្វក្នុងសត្វល្អិតមានសមត្ថភាពហោះហើរ បានឆ្ងាយឆ្ងាយក្នុងការស្វែងរកស្រូវជាច្រើនបានដោយងាយស្រួល និងធ្វើឱ្យការស្វែងរកសត្វល្អិតនិងការបង្កើនចំនួនកូនបានយ៉ាងលឿន។ គុណសម្បត្តិនៃសត្វល្អិតគឺជាច្រើនគឺ ពួកវាអាចបង្កប់ស្រទាប់ស្របតាមផងខ្លួនរបស់វាឱ្យ ការនេះផ្តល់ឱ្យពួកវា ផ្លែបន្តជីវិតបានច្រើនជាងសត្វល្អិតដទៃទៀត។ សត្វល្អិត ផ្លែបន្តជីវិតបានច្រើនជាងសត្វល្អិតដទៃទៀត។ សត្វល្អិត គឺជាសត្វក្នុងចំណោមសត្វល្អិតដែលមានចំណេះដឹងខ្ពស់បំផុត។	Q1. សត្វក្នុងសត្វល្អិតក្នុងការបង្កប់ស្រទាប់ស្របតាមផងខ្លួនបង្កើនមួយណាមួយខាងក្រោម?			
		1. ការផ្គត់ផ្គង់ស្បៀងអាហារ	2. កន្លែងលាក់ខ្លួន	3. ការស្វែងរកការបង្កើន	4. ល្បឿនហោះហើរ
Portuguese por_Latn	Insetos foram os primeiros animais a ir para o ar. Sua habilidade de voar os ajudou a fugir mais facilmente de inimigos e a encontrar alimento e parceiros de maneira mais eficiente. A maioria dos insetos têm a vantagem de poder dobrar suas asas ao longo do corpo. Isso lhes dá uma gama maior de pequenos locais para se esconderem de predadores. Hoje, os únicos insetos que não podem dobrar suas asas para trás são libélulas e efêmeras.	Q1. A capacidade de um inseto de dobrar suas asas para trás aumenta qual das seguintes opções?			
		1. Suprimento de comida	2. Esconderijos	3. Encontrar parceiros/as	4. Velocidade de voo
		Q2. Qual das seguintes opções não é mencionada no trecho como algo afetado pela habilidade de voar de um inseto?			
		1. Reprodução	2. Vida útil	3. Obtenção de alimentos	4. Eficiência

Figure 1: A sample passage from the dataset in 4 different languages, displayed alongside its two questions.

ately punish biased models. The English questions on their own present a significant challenge to numerous models, while humans are capable of answering the questions with near-perfect accuracy.

The first of its scale, BELEBELE is parallel across all languages, facilitating a direct comparison of model performance across all languages. The dataset covers typologically diverse languages across 29 scripts and 27 language families. Seven languages are included in two separate scripts, resulting in one of the first NLP benchmarks for the romanized variants of Hindi, Urdu, Bengali, Nepali, and Sinhala. We further detail our data collection process and the resulting corpus in Section 3.

The dataset enables evaluation of mono- and multi-lingual models, but the parallel nature also enables a number of cross-lingual evaluation settings. We evaluate several masked language models (MLMs) after fine-tuning on an English training set as well as with the assistance of machine translation (Translate-Train-All). For LLMs, we evaluate several models using In-Context Learning and also instruction-tuned models via Zero-Shot. We discuss our results in Section 5.

2 Background

2.1 Cross-Lingual Evaluation Benchmarks

There are several datasets for NLU that are parallel across numerous languages and enable monolingual, multilingual, or cross-lingual evaluation. These include XNLI (Conneau et al., 2018),

XQUAD (Artetxe et al., 2020), and MLQA (Lewis et al., 2020). MINTAKA (Sen et al., 2022) is designed with LLMs in mind, presenting a more difficult QA task in 9 languages. Beyond QA, XL-SUM (Hasan et al., 2021) is an analogous dataset in the domain of abstractive summarization. However, all these datasets together cover under 30 languages, most of which are high- or medium- resource. MASSIVE (FitzGerald et al., 2023) is a large NLU dataset covering 51 languages, but in the domain of spoken conversational agents. NER (Pan et al., 2017) has extensive language coverage and TYDIQA (Clark et al., 2020) is a popular multilingual benchmark but neither are parallel.

Our work undertakes the challenge of expanding existing cross-lingual evaluations to 122 languages, many of which currently lack any NLU benchmark at all.

2.2 Non-English Machine Reading Comprehension

While the question-answering portion varies, machine reading comprehension (MRC) tasks are defined by the closed-book passage provided to answer each question. Of course, a big majority of MRC datasets are in English, such as TRIVIAQA (Joshi et al., 2017) and the BABI tasks (Weston et al., 2016).

However, the need for MRC datasets for other languages has led to a proliferation of monolingual closed-book MRC datasets in recent years

BELEBELE Statistics					
Languages	Passage statistics			Question statistics	
Total Number	122	Distinct Passages	488	Distinct Questions	900
Distinct Languages (ignoring script)	115	Questions per passage	1-2	Multiple-choice answers (num correct) per question	4 (1)
Language Families	27	Avg. words per passage (std)	79.1 (26.2)	Avg. words per question (std)	12.9 (4.0)
Scripts	29	Avg. sentences per passage (std)	4.1 (1.4)	Avg. words per answer (std)	4.2 (2.9)

Table 1: Language and Text Information for BELEBELE. Average length statistics are computed on the English split.

(Mozannar et al., 2019; Hardalov et al., 2019; d’Hoffschmidt et al., 2020; Möller et al., 2021; Anuranjana et al., 2019; Gupta et al., 2018; Croce et al., 2018; Efimov et al., 2020; Shavrina et al., 2020; Sun et al., 2021). Most were created using translation and so are parallel with an English dataset, often SQUAD (Rajpurkar et al., 2016). However, BELEBELE aims to cover these languages and more in one consistent dataset.

2.3 Multiple Choice QA

Compared to extractive QA, multiple-choice is a less common form of MRC datasets. Some, like RACE (Lai et al., 2017), are made from exam questions for English learners, while others were built specifically for NLU systems, like MCTest (Richardson et al., 2013) and MultiRC (Khashabi et al., 2018). While most are intended to be closed-book, SCIQ (Welbl et al., 2017) and OPEN-BOOKQA (Mihaylov et al., 2018) require open information retrieval. Others, like COPA (Roemmele et al., 2011), SWAG (Zellers et al., 2018), and RECLOR (Yu et al., 2020), require higher-level commonsense reasoning to answer. For multilingual systems, EXAMS (Hardalov et al., 2020) is a parallel multiple-choice QA dataset covering 28 languages. However, no passages are provided and answering questions requires cross-lingual knowledge transfer and reasoning.

2.4 FLORES-200

The passages in the BELEBELE corpus are directly sourced from the FLORES-200 Machine Translation Benchmark (Goyal et al., 2022; NLLB et al., 2022). The parallel dataset was constructed by sourcing English passages from Wikinews, Wikijunior, and WikiVoyage. The translations were performed by native speakers with high English fluency and translation experience. Translators were instructed to maintain informative and standardized content while handling named entities, abbreviations, idioms, and pronouns appropriately.

3 The BELEBELE Dataset

We opted to create multiple-choice questions and answers in English and then translate, as opposed to

creating resources natively in each language. Many of the advantages to this approach outlined in Conneau et al. (2018) remain. Most importantly, this leads to significantly more similar sets of samples across languages, enabling direct score comparison. The process for creating the dataset is summarized in Figure 2.

3.1 Creation of Multiple Choice Questions & Answers

To create the BELEBELE dataset, we first construct a question-answering dataset in English.

Amongst machine reading comprehension tasks, we select multiple-choice questions (MCQs) because it would lead to the fairest evaluation across languages. Related tasks, such as span extraction, are more sensitive to morphological differences, making scaling to many languages difficult (Lewis et al., 2020). In addition, MCQs enable us to better center the questions on information explicitly stated in the passage, as yes/no or entailment (NLI) questions can be easier to answer with external knowledge held in pretrained models. In order for the questions to discriminate solely between different levels of language comprehension, we intentionally create questions that do not require higher levels of information processing, such as multi-hop or commonsense reasoning.

Constructing high quality MCQs depends most importantly on creating strong negatives that are neither obviously wrong nor possibly correct (Agarwal and Mannem, 2011; Richardson et al., 2013). We do not want the dataset to be easy enough for biased models (e.g. models that use shortcuts or pattern-match) (Boyd-Graber and Börschinger, 2020). In setting up this annotation, we consider the protocols proposed in Bowman et al. (2020) and the warnings from Malaviya et al. (2022). We implement an iterative procedure with the Language Service Provider (LSP) for this involved data collection task, similar to that from Nangia et al. (2021). We engaged in 5 total iterations, providing and receiving feedback in each. Annotators were instructed on the similarities and differences on how ML models approach QA datasets versus humans, which we felt substantially improved the

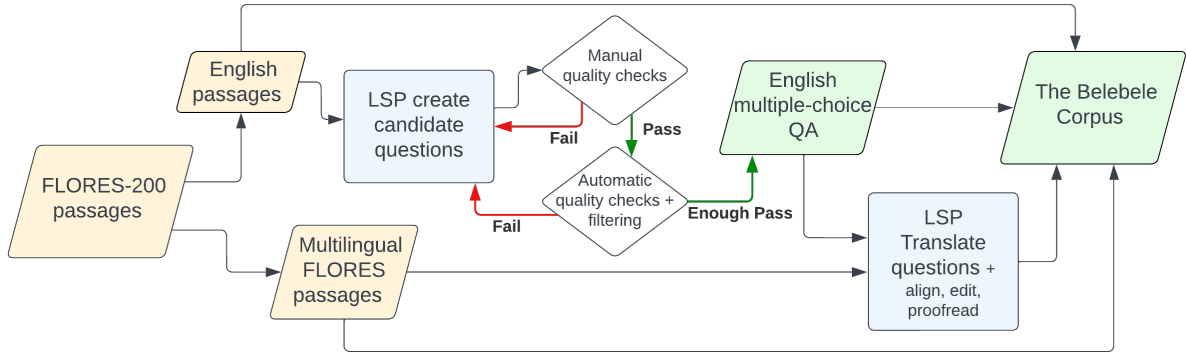


Figure 2: Illustration of the dataset creation process with Language Service Provider (LSP) starting from FLORES

quality of the data.

Our final guidelines include both important points such as having the correct response being unambiguous, as well as particularized rules such as *no double negatives* (Mihaylov et al., 2018). For each rule we provided annotators with a good and bad example to illustrate. An abridged version of our guidelines can be found in the Appendix A.5.1.

3.2 Quality Assurance

At each iteration, we evaluate whether or not returned samples satisfy the minimum quality bar through a mix of manual inspection and automatic inspection. At every step, we manually verified a sample of questions to understand how well the annotators were on the same page with us about the guidelines. While time consuming, manual verification was the most assured way to provide tangible feedback to the annotators, notably on the difficulty of the questions created. As we progressively aligned with annotators, we were required to look over more samples to provide feedback.

To complement the manual inspection of a subset of questions, we use programmatic methods to evaluate all questions from a statistical perspective. Based on the findings in Malaviya et al. (2022), we create low-level features to identify overly easy questions or low-effort strategies employed by annotators. For example, we evaluate the lexical overlap between different combinations of the texts associated with a question to evaluate whether the question is answerable by a biased model. This allows us to see if the question can be answered without the passage, without the question, or with only one sentence in the passage. We also identified patterns associated with heuristic solvability, such as the wrong answers less frequently being extracted from the passage. We detail these features in Appendix A.6.

These low-level features allow us to (1) determine whether an annotation iteration was up to par, (2) filter out questions that failed these heuristic checks (for the final iteration, about 20% were filtered out), and (3) compare to other MCQ datasets. We run statistical t-tests to ensure the distribution of these features for correct answers is no different than for wrong answers. In comparison to MCTEST which largely fails this t-test (p-value < 0.01), our final collection has p-value 0.81. We also train a logistic regression model to answer using only bag-of-word representations and find that the best the naïve model could achieve was an accuracy of 0.28 on our 900 questions. This is just better than random (0.25) and much lower than what was achieved on MCTEST, 0.44.

3.3 Translating the Corpus

BELEBELE was created end-to-end without the use of machine translation technology, relying solely on experts fluent in English and the target language.

For all languages included in the corpus, the context passages were taken directly from the FLORES-200 dataset, with the exception of Hindi, Bengali, Urdu, Nepali, and Sinhala in the Latin script. While the romanized variant of these 5 Indo-Aryan languages is very prevalent on the modern Internet, their romanization is not included in FLORES-200. We thus had annotators transliterate from the native to Latin script with the support of IndicXlit (Madhani et al., 2023). As a result, much like Modern Standard Arabic, these languages are present in two forms in the corpus.

In order for the questions and answers to properly pair the translated FLORES passages, the latter was provided for the annotators. We specifically instructed annotators to align potentially ambiguous translations with the original passages. While Clark et al. (2020) warns that this forced alignment

Model	Size/Variant	Vocab size	AVG	% ≥ 50	% ≥ 70	eng_Latn	non-Eng AVG
<i>5-Shot In-Context Learning (examples in English)</i>							
LLAMA 1	7B	32K	27.7	0.0%	0.0%	37.3	27.6
LLAMA 1	13B	32K	30.4	0.8%	0.0%	53.3	30.2
LLAMA 1	30B	32K	36.2	18.0%	0.8%	73.1	35.9
LLAMA 1	70B	32K	40.9	25.4%	12.3%	82.5	40.5
LLAMA 2 base	70B	32K	48.0	38.5%	26.2%	90.9	47.7
FALCON	40B	65K	37.3	16.4%	1.6%	77.2	36.9
<i>Zero-Shot for Instructed Models (English instructions)</i>							
BLOOMZ**	7.1B	251K	43.2	28.7%	9.0%	79.6	42.9
LLAMA-2-CHAT	7B	32K	34.4	4.1%	0.0%	58.6	34.1
LLAMA-2-CHAT	70B	32K	41.5	27.0%	2.5%	78.8	41.2
GPT3.5-TURBO	unk	100K	51.1	44.2%	29.2%	87.7	50.7
<i>Full Finetuning in English</i>							
XLM-R	large (550M)	250K	54.0	64.8%	15.6%	76.2	53.8
XLM-V	large (1.2B)	902K	55.6	69.7%	21.2%	76.2	54.9
INFOXML	large (550M)	250K	56.2	67.2%	28.7%	79.3	56.0
<i>Translate-Train-All</i>							
XLM-R	large (550M)	250K	58.9	69.7%	36.1%	78.7	58.8
XLM-V	large (1.2B)	902K	60.2	76.2%	32.8%	77.8	60.1
INFOXML	large (550M)	250K	60.0	70.5%	36.9%	81.2	59.8

Table 2: Summary of results on BELEBELE across models and evaluation settings. % $\geq 50/70$ refers to the proportion of the 122 languages for which a given model performs above 50/70%. We additionally report LLAMA-2-CHAT zero-shot results leveraging translation in Table 3. ** In this table, we present results from BLOOMZ despite it being finetuned for translation with FLORES data, presenting an unfair advantage for the Zero-Shot setting (See our Ethics Statement).

could increase ‘translationese’, it is necessary to ensure equivalent question difficulty across languages. The modifications to the translation guidelines can be found in Appendix A.5.2. All translations were proofread and edited by an additional annotator.

3.4 English Training Data

BELEBELE is intended to be used as a test set, and not for training. Therefore, for models that require additional task finetuning, we instead assemble a training set consisting of samples from English multiple-choice QA datasets (See Appendix A.2).

3.5 The BELEBELE Dataset in Summary

BELEBELE contains 900 questions, each with 4 multiple-choice answers and one correct answer. Most passages have two associated questions, but some have only one. In total, there are 488 distinct passages, none belonging to the hidden FLORES test set. Parallel across 122 languages, the corpus contains a total of 109,800 rows. Amongst the language varieties, there are 29 unique scripts and 27 language families represented (see Figure 4). Some text and language statistics are displayed in Table 1 and we display a sample passage in four languages in Figure 1.

Because of the careful annotation procedure and

quality checks, the MCQs discriminate text comprehension competence. It often includes paraphrasing and strong negatives in order to elude simple pattern-matching models. Questions often additionally require understanding multiple sentences. However, answering does not require presumptions or external knowledge as is required in more difficult reasoning datasets. For example, Q1 in Figure 1 is unambiguous. *Food*, *mates*, and *flying* are all mentioned in the passage, but a careful read reveals the wings folding back is only associated with *hiding spaces*. To confidently rule out the other candidate answers, it is required to understand three sentences. In general, we find all questions to be answerable by humans fluent in the target language, but not without focused reading (see Section 5.1).

As can be seen in Figure 1, the passages, questions, and answers are aligned in semantic meaning and formality. BELEBELE therefore poses an equivalent challenge in all languages. It also enables models with cross-lingual alignment in the semantic representation space to answer questions when passage, question, and answers are swapped to different languages. Since FLORES includes passages in 83 additional languages, we can even evaluate reading comprehension in these languages

by asking the questions in English.

4 Experiments

Thanks to BELEBELE, we are able to evaluate numerous models and establish baseline performances across 122 language variants. We compare performance between popular multilingual MLMs and LLMs in several settings. For all, accuracy is the central metric. With 4 candidate answers for each question, the expected accuracy for sequence classification models that guess randomly is 0.25.³

4.1 Evaluated Models

Masked Language Models (MLMs) We evaluate three different models, XLM-V (Liang et al., 2023), INFOXLM (Chi et al., 2021), and XLM-R (Conneau et al., 2020a). All the evaluated MLMs have been pretrained on intentionally multilingual corpora inclusive of about 100 languages. The pre-training data in high-resource languages is typically down-sampled while low-resource languages are up-sampled in order to favor multilingual performance (Conneau et al., 2020a). In addition, all their subword tokenizers (Kudo and Richardson, 2018) are trained on multilingual corpora, making them better suited for multilingual text.

Large Language Models We evaluate GPT3.5-TURBO, FALCON, and LLAMA (1 and 2). GPT3.5-TURBO is a model optimized for chat based on GPT-3 (Brown et al., 2020) available through OpenAI APIs⁴. Limited details have been disclosed about the pretraining and fine-tuning data.⁵ LLAMA 1 (Touvron et al., 2023a) is a collection of decoder-only transformers models trained on 1T (for 7B, 13B) or 1.4T (for 30B, 65B) tokens of publicly available online data, while LLAMA 2 (Touvron et al., 2023b) is pretrained on about 2T. We evaluate all four pretrained checkpoints for LLAMA 1. We evaluate both LLAMA 2 70B’s pretrained version and its chat version instruction-fine-tuned for safe dialog purposes (a.k.a. LLAMA-2-CHAT). We also evaluate FALCON 40B, which was pretrained on 1T extensively filtered web-crawled samples (Penedo et al., 2023).

While LLAMA 1 was reportedly trained in 20 languages with Latin and Cyrillic scripts, non-English text accounts for less than 4.5% of the pretraining

corpus (Touvron et al., 2023a). LLAMA 2 pretraining data is made of 89.7% of English data, 8.4% unidentified, and a tiny 1.9% belonging to 26 other languages⁶ (Touvron et al., 2023b). Both series use the same BPE-based tokenizers (Kudo and Richardson, 2018). Splitting unicode characters into bytes also helps LLAMA avoid out-of-vocabulary errors.

In Table 2, we present results for BLOOMZ-7B (Muennighoff et al., 2023), which was both pre-trained and instruction finetuned on heavily multilingual data and has significant vocabulary capacity. However, we do not discuss it in our analyses, as it was fine-tuned for translation on FLORES-200, undermining fair assessment.

4.2 Evaluation Settings

More specifics are provided in Appendix A.4.

Full Model Fine-tuning For evaluating MLMs, we add a multiple-choice classification head and fine-tune the entire model. We finetune in two settings, (1) in English and evaluate zero-shot cross-lingual transfer and (2) on machine-translated samples of the training set to all the target languages and evaluate each language (*Translate-Train-All*).

Five-Shot In-Context Learning We evaluate the pretrained LLAMA 1 and 2 as well as FALCON 40B in the five-shots setting. Examples are sampled from the English training set and prompted to the model. For prediction, we pick the answer with highest probability and report the average cumulative score over 3 runs.

Zero-Shot Evaluation We evaluate GPT3.5 and LLAMA-2-CHAT (70B) in Zero-Shot by describing the task with natural language instructions (in English). We present the passage, question, and four possible answers, and instruct the model to provide the letter of the answer. We post-process answers and accept multiple formats.⁷

In addition, we prompt LLAMA-2-CHAT with instructions that are machine translated to the target language from English. Conversely, we evaluate machine-translating the passages, questions, and answers back to English and prompting them to the model (*Translate-Test*). This setting allows us to compare in-language comprehension to the popular approach of cascading with machine translation.

³Note: For sequence-to-sequence models (e.g. instructed models) that are evaluated in exact-match scenarios, this lower-bound does not hold.

⁴<https://platform.openai.com/docs/models>

⁵Our analyses rely on the unverified assumption that GPT3.5-TURBO was not trained on FLORES. See Limitations

⁶See Table 10 in Touvron et al. (2023b) for a full list of the identified languages

⁷Code provided at github.com/facebookresearch/belebele

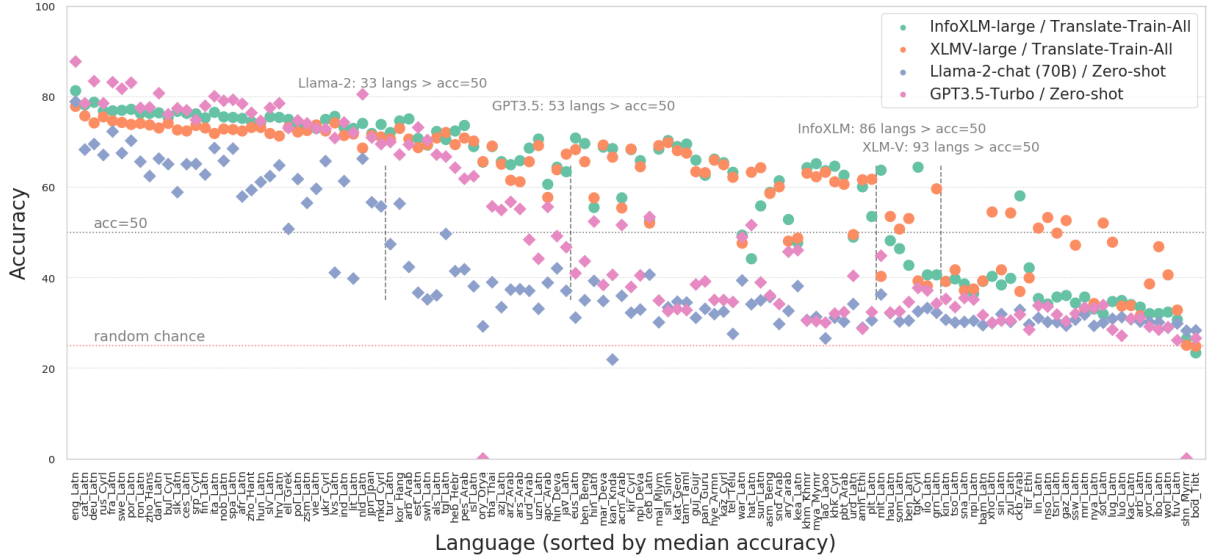


Figure 3: BELEBELE Results in 122 languages. We compare four models in two settings and see the difference between intentionally multilingual models and models with English-centric data. GPT3.5-TURBO performs the best on the top 20 languages, but after 40-50, its performance falls far behind INFOXLM and XLM-V. Similarly, INFOXLM outperforms XLM-V in the first 40 languages, but XLM-V proves more capable on the long tail of languages. Note that the language order can change the plot considerably, here we choose median accuracy.

5 Results

We provide summarized results in Table 2 and detailed results in Appendix A.7.

5.1 How difficult is BELEBELE?

As discussed in Section 3, the questions in BELEBELE are intentionally difficult. While the primary challenge of this dataset is its multilinguality, we see that empirically, the English questions are able to shed light on the varying NLU capabilities of models. With full finetuning, we achieved a maximum accuracy of 71.7 in English with ROBERTA-base model, significantly less than the 90.9 achieved by LLAMA 2 70B in five-shot. Between LLAMA 1 variants, we see a wide range of results, with the 7B model only achieving 37.3. So while the primary difficulty of BELEBELE is its multilinguality, we see a wide range of performance between different model types and sizes.

In addition, all models evaluated comfortably underperform humans. To establish human performance, 4 authors each randomly sampled around 30 English MCQs and answered in a blind test, achieving mean 97.6 accuracy⁸. This is much higher than any of the models evaluated, implying the task presents a particular challenge for models and that there is room to improve. For comparison, Nangia and Bowman (2019) conservatively estimate human performance to be 92.8 on the English

split of XNLI (i.e. MNLI (Williams et al., 2018)).

When comparing model performance with XNLI, we find very high correlation. In the Translate-Train-All setting, XLM-V, INFOXLM, and XLM-R all perform about 10 accuracy points lower on BELEBELE than on XNLI Translate-Train⁹ reported in Liang et al. (2023) and Chi et al. (2021). Still, across all 15 languages and three models, we find a score correlation of $r = 0.85$.

5.2 Multilingual Generalization of MLMs and LLMs on BELEBELE

Schematically, the performance of a language model in a given language is related to two key factors. (i) First, the amount of pretraining data in the target language. As predicted by the scaling laws (Kaplan et al., 2020), performance in a language increases monotonically with the amount of pretraining tokens. (ii) Second, the cross-lingual transfer happening between languages in the pretraining data and the target language at inference time (Conneau et al., 2020a,b). This transfer is impacted by a combination of typological, script, and lexical similarities between the pretraining languages and the target language (Muller et al., 2021a, 2023). These two factors are hard to disentangle due to the scale (up to ~ 1 T tokens) and the potential language leaks of large-scale pretraining corpora (Kreutzer et al., 2022). Thanks to BELEBELE’s quality and scale,

⁸95% CI for all 900 questions = [93.1, 99.5]

⁹In traditional Translate-Train, the model is finetuned on translated training inputs for each language *individually*.

Model	Variant	Eval Setting	AVG	% ≥ 50	% ≥ 70	eng_Latn
<i>Translate-Test (English) on 91 non-English languages in Zero-Shot</i>						
LLAMA-2-CHAT	70B	Translate-Test	57.1	78.0%	2.2%	78.8
LLAMA-2-CHAT	70B	In-Language	44.1	35.2%	2.2%	78.8
<i>Translated Instructions in 89 non-English languages Zero-Shot</i>						
LLAMA-2-CHAT	70B	In-Language Translated Instructions	38.7	36.0%	7.9%	78.8
LLAMA-2-CHAT	70B	English Instructions	44.9	37.1%	3.4%	78.8

Table 3: Results of LLAMA-2-CHAT in zero-shot in two machine translation-based evaluation settings; Translate-Test (passages, questions, answers translated back to English) and evaluations with the English instructions translated to the target language. The traditional setting on the same languages is provided for comparison. Note that the summary statistics differ from Table 2 because this is for a subset of all languages on which we performed these translation-based evaluations.

we provide detailed evidence of both impacting the multilingual generalization of the models.

Impact of Pretraining Language Distribution

One of the key differences between the MLMs and LLMs evaluated is their pretraining data distribution and parameter size, explaining the large performance differences between them. For instance, LLAMA 2 largely outperforms XLM-R on high-resource languages, but only achieves accuracy 50 on about half the amount of languages as XLM-R (See Table 2). This difference between the MLMs and LLMs evaluated is illustrated in Figure 3. However, despite this gap, all LLMs evaluated perform surprisingly well on a large number of languages. For instance, LLAMA-2-CHAT is above 35 accuracy (i.e. 10 above random) for 59 languages. This shows that English-centric LLMs are a promising starting point to build multilingual models.

Machine Translation for Zero-Shot Our Translate-Test evaluations show that using machine translation into English strongly outperforms LLAMA-2-CHAT (70B) performance in the original target language. Across 91 evaluated languages, only 2 are non-trivially better in-language (German and Italian), compared to 68 (none high-resource) for which translating to English is better, none of which are considered high-resource. Compared to LLAMA-2-CHAT having zero-shot accuracy above 50% for 33 languages, it has 71 in Translate-Test (see Appendix A.7.4).

In addition, we evaluate machine-translating the task instructions to the target language. For around 25 languages, the translated instructions were not well understood (i.e. accuracy less than random), correlating strongly with already low-score languages. For the rest, the performance relative to using English instructions is mixed, though lan-

guages that already scored highly had the largest accuracy boost from in-language instructions. While machine-translating instructions are less effective than quality in-language instructions, these results do not suggest that the use of English instructions is the primary reason for why performance on non-English languages lags English so significantly.

Impact of Sub-Word Tokenization We reaffirm a correlation between increasing vocabulary size and performance on lower resource languages (Liang et al., 2023). XLM-V has a massive 900k-token vocabulary that allocates capacity for each individual language and de-emphasizes token sharing between languages. XLM-V outperforms XLM-R and INFOXML (250k vocabulary size) on low-resource languages even though they all have the same architecture and are trained on the same dataset (CC-100). GPT3.5-TURBO (100k vocabulary size),¹⁰ FALCON (65k vocabulary size), and LLAMA 2 (32k vocabulary size) all fall off abruptly for medium- and low- resource languages. Larger vocabulary size may explain why FALCON 40B performs equivalent to LLAMA 1 30B despite having been pretrained on fewer non-English tokens.

Scaling effect on Multilingual Generalization

We report in Figure 4 the impact of model sizes on performance on the BELEBELE benchmark across six language families and English. We find that scale is critical for LLAMA 1 to perform reading comprehension as the 7B checkpoint performs slightly above chance in English even. As the parameter size increases, performances across the board increases significantly. Only the 30B and 65B checkpoints are able to perform non-trivially in language families not reported to be in the pre-training corpus (Japanese and Greek). However, un-

¹⁰According to <https://github.com/openai/tiktoken>.

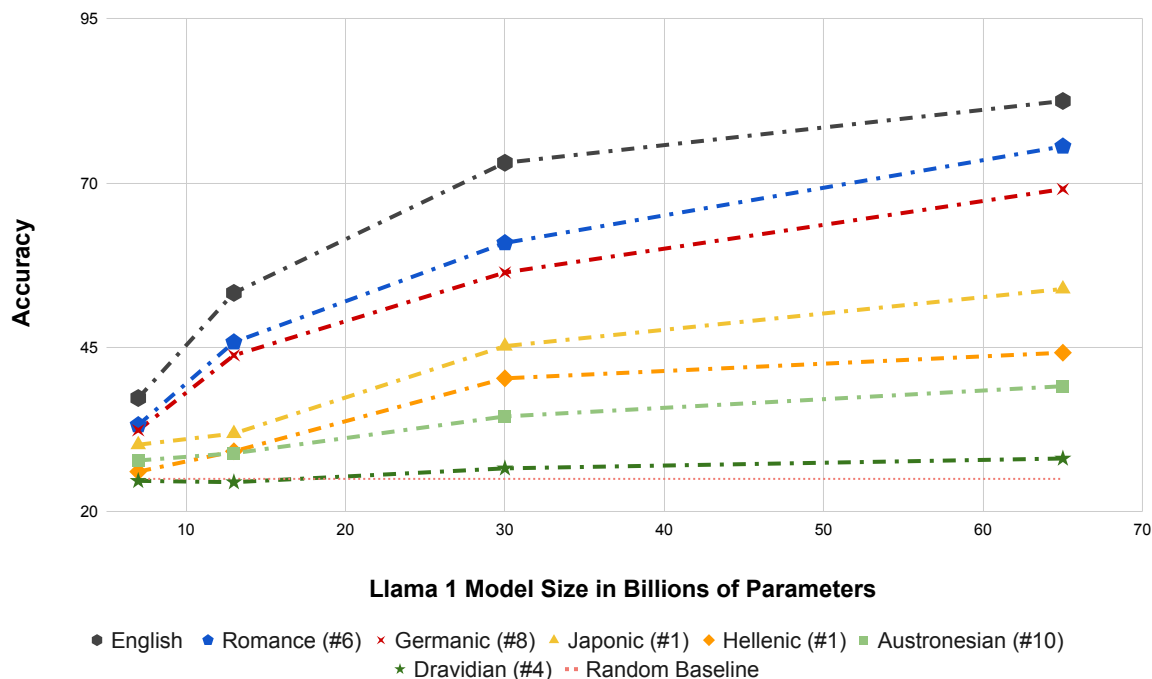


Figure 4: Impact of Models’ scale (from 7B to 65B parameters of LLAMA 1) on the performance on BELEBELE for 6 language families and English. The number of languages in a given family is indicated as (#N). LLAMA 1 is evaluated in the 5-shot settings with examples from the training data in English. Scores are averaged over 3 runs.

like other language families such as Romance and Germanic, the performance becomes non-trivial only with the 30B and 65B checkpoints. Results like this suggest that generalizing to distant languages after English-centered pretraining requires more parameters.

Impact of Script Comparing the Romanized versions with the original scripts for Hindi, Urdu, Bengali, Sinhala, Nepali, and Modern Standard Arabic, we find that all models except FALCON perform stronger in the native script than in the Latin script (see Appendix A.7.3). However, the native scripts are allegedly not present in the pretraining data for LLAMA 2 and FALCON. For the Indo-Aryan languages, we hypothesized cross-lingual transfer would be higher in the Latin variant since the tokenization will be more suitable and there is opportunity for shared subwords (anchor points) (Conneau et al., 2020b; Muller et al., 2020; Pfeiffer et al., 2021; Muller et al., 2021a). However, this only seems to be the case for FALCON. The results generally suggest the models were pretrained on significant samples in the native script (perhaps due to code-switching or poor language identification).

6 Conclusion

A fundamental limitation to conducting sound evaluations of the capabilities of language models in

low-, or even moderate-, resource languages is the availability of annotated benchmarks. This paper presents a massive dataset, BELEBELE, consisting of passages and multiple-choice questions evaluating reading comprehension in 122 languages. This benchmark enables critical evaluation of reading comprehension capabilities of LLMs in English and top languages. In addition, the dataset is the first of its kind in many medium- and low-resource languages, enabling unprecedented insight into the multilingual capabilities of language models. We present results from a number of popular MLMs and LLMs in different evaluation settings and find that while large vocabulary size and balanced pretraining data correlates with highest model performance on medium- and low-resource languages, even English-centric LLMs can go a long way and generalize to over 30 languages.

For future work, we hope the many evaluations and experiments now possible will allow for deeper dives into current language models. BELEBELE can also complement investigations into specific model capabilities (e.g. reasoning), leading to a broader understanding of the relationship between such abilities and multilinguality. As a result, we believe BELEBELE will soon unveil further insights that contribute to the development of NLP systems beyond high-resource languages.

Limitations

Pretraining Documentation Our model analyses are limited by inconsistent documentation on the composition of pretraining corpora used. We present results from BLOOMZ (Muennighoff et al., 2023) only in very limited fashion because its documentation showed it was fine-tuned for translation on FLORES-200. However, as alluded to in Section 4.1, we present GPT3.5-TURBO results on BELEBELE even though we cannot verify its pretraining or finetuning data. Because of this, comparing results on GPT3.5-TURBO to other models may be unfair given the lack of transparency on training data. To enable further understanding of the interaction of multilingual text during training, we point to two critical research directions to enable progress in the field. First, (i) better language identification systems: popular language identification models are trained on a restricted number of languages and domains and only work at the sentence level (Bojanowski et al., 2017), limiting their abilities to track languages in code-switched data and embedded text. Second, (ii) we encourage LLM developers to improve reporting on pretraining language distribution. This is necessary for the research community to understand the cross-lingual transfer capabilities of LLMs and to improve NLP system design for low-resource languages.

Errors in FLORES As briefly mentioned in Section 3.3, annotators discovered a few quality issues with the FLORES translations (i.e. the original annotations done prior to this work). Some of them are likely due to style/dialect differences between annotators, but many seem to not be. It's rare enough, thanks to the extensive quality-assurance loops implemented by the NLLB team and the LSP. However, over the scale of 122 languages a fair number of issues have arisen, especially in lower-resource languages. Since updating the base FLORES dataset is not in scope for this project, we deliberated on each with the LSP to maximize both appropriateness and cross-language consistency of the question/answers translations.

Translationese Even with our extensive quality assurance, we warn that "translationese" may change the nature of the task across languages. In numerous cases, the *perfect* translation does not exist. This may cause accuracy on non-English languages to not be directly comparable to on English itself.

Ethics Statement

Open-Source Our decision to open-source BELEBELE may compromise future benchmarking as the samples may get collected into large pretraining corpora, undermining fair comparison. This is especially the case for zero- or few-shot evaluation. Nonetheless, we consciously determine the value of open-sourcing the full dataset to far outweigh these considerations.

English-centrism BELEBELE was designed to measure the reading comprehension abilities of NLP systems across 122 languages. We specifically align as much as possible with translation choices made in the creation of FLORES. Therefore, by-design the samples collected do not capture language- and culture-specific phenomena such as formality (Ersoy et al., 2023), values (Kovač et al., 2023), and aboutness (Hershcovich et al., 2022). While conscious of this Western-centrism, BELEBELE was designed to prioritize comparability across languages. Following BELEBELE, building NLP systems inclusive of all cultures and languages will require the release of benchmarks that capture these phenomena.

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A Appendix

A.1 Languages and Variants

Language Code	Name in English	Script	Family
acm_Arab	Mesopotamian Arabic	Arab	Afro-Asiatic
afr_Latn	Afrikaans	Latn	Germanic
als_Latn	Tosk Albanian	Latn	Paleo-Balkan
amh_Ethi	Amharic	Ethi	Afro-Asiatic
apc_Arab	North Levantine Arabic	Arab	Afro-Asiatic
arb_Arab	Modern Standard Arabic	Arab	Afro-Asiatic
arb_Latn	Modern Standard Arabic (Romanized)	Latn	Afro-Asiatic
ars_Arab	Najdi Arabic	Arab	Afro-Asiatic
ary_arab	Moroccan Arabic	Arab	Afro-Asiatic
arz_Arab	Egyptian Arabic	Arab	Afro-Asiatic
asm_Beng	Assamese	Beng	Indo-Aryan
azj_Latn	North Azerbaijani	Latn	Turkic
bam_Latn	Bambara	Latn	Mande
ben_Beng	Bengali	Beng	Indo-Aryan
ben_Latn	Bengali (Romanized)	Latn	Indo-Aryan
bod_Tibt	Standard Tibetan	Tibt	Sino-Tibetan
bul_Cyrl	Bulgarian	Cyrl	Balto-Slavic
cat_Latn	Catalan	Latn	Romance
ceb_Latn	Cebuano	Latn	Austronesian
ces_Latn	Czech	Latn	Balto-Slavic
ckb_Arab	Central Kurdish	Arab	Iranian
dan_Latn	Danish	Latn	Germanic
deu_Latn	German	Latn	Germanic
ell_Grek	Greek	Grek	Hellenic
eng_Latn	English	Latn	Germanic
est_Latn	Estonian	Latn	Uralic
eus_Latn	Basque	Latn	Basque
fin_Latn	Finnish	Latn	Uralic
fra_Latn	French	Latn	Romance
fuv_Latn	Nigerian Fulfulde	Latn	Atlantic-Congo
gaz_Latn	West Central Oromo	Latn	Afro-Asiatic
grn_Latn	Guarani	Latn	Tupian
guj_Gujr	Gujarati	Gujr	Indo-Aryan
hat_Latn	Haitian Creole	Latn	Atlantic-Congo
hau_Latn	Hausa	Latn	Afro-Asiatic
heb_Hebr	Hebrew	Hebr	Afro-Asiatic
hin_Deva	Hindi	Deva	Indo-Aryan
hin_Latn	Hindi (Romanized)	Latn	Indo-Aryan
hrv_Latn	Croatian	Latn	Balto-Slavic
hun_Latn	Hungarian	Latn	Uralic
hye_Armn	Armenian	Armn	Armenian
ibo_Latn	Igbo	Latn	Atlantic-Congo
ilo_Latn	Ilocano	Latn	Austronesian
ind_Latn	Indonesian	Latn	Austronesian
isl_Latn	Icelandic	Latn	Germanic
ita_Latn	Italian	Latn	Romance
jav_Latn	Javanese	Latn	Austronesian
jpn_Jpan	Japanese	Jpan	Japonic
kac_Latn	Jingpho	Latn	Sino-Tibetan
kan_Knda	Kannada	Knda	Dravidian
kat_Geor	Georgian	Geor	Kartvelian
kaz_Cyrl	Kazakh	Cyrl	Turkic
kea_Latn	Kabuverdianu	Latn	Portuguese Creole
khk_Cyrl	Halh Mongolian	Cyrl	Mongolic
khm_Khmr	Khmer	Khmr	Austroasiatic
kin_Latn	Kinyarwanda	Latn	Atlantic-Congo
kir_Cyrl	Kyrgyz	Cyrl	Turkic
kor_Hang	Korean	Hang	Koreanic
lao_Lao	Lao	Lao	Kra-Dai
lin_Latn	Lingala	Latn	Atlantic-Congo
lit_Latn	Lithuanian	Latn	Balto-Slavic
lug_Latn	Ganda	Latn	Atlantic-Congo
luo_Latn	Luo	Latn	Nilo-Saharan
lvs_Latn	Standard Latvian	Latn	Balto-Slavic

mal_Mlym	Malayalam	Mlym	Dravidian
mar_Deva	Marathi	Deva	Indo-Aryan
mkd_Cyrl	Macedonian	Cyrl	Balto-Slavic
mlt_Latn	Maltese	Latn	Afro-Asiatic
mri_Latn	Maori	Latn	Austronesian
mya_Mymr	Burmese	Mymr	Sino-Tibetan
nld_Latn	Dutch	Latn	Germanic
nob_Latn	Norwegian Bokmål	Latn	Germanic
npi_Deva	Nepali	Deva	Indo-Aryan
npi_Latn	Nepali (Romanized)	Latn	Indo-Aryan
nso_Latn	Northern Sotho	Latn	Atlantic-Congo
nya_Latn	Nyanja	Latn	Afro-Asiatic
ory_Orya	Odia	Orya	Indo-Aryan
pan_Guru	Eastern Panjabi	Guru	Indo-Aryan
pbt_Arab	Southern Pashto	Arab	Indo-Aryan
pes_Arab	Western Persian	Arab	Iranian
plt_Latn	Plateau Malagasy	Latn	Austronesian
pol_Latn	Polish	Latn	Balto-Slavic
por_Latn	Portuguese	Latn	Romance
ron_Latn	Romanian	Latn	Romance
rus_Cyrl	Russian	Cyrl	Balto-Slavic
shn_Mymr	Shan	Mymr	Kra-Dai
sin_Latn	Sinhala (Romanized)	Latn	Indo-Aryan
sin_Sinh	Sinhala	Sinh	Indo-Aryan
slk_Latn	Slovak	Latn	Balto-Slavic
slv_Latn	Slovenian	Latn	Balto-Slavic
sna_Latn	Shona	Latn	Atlantic-Congo
snd_Arab	Sindhi	Arab	Indo-Aryan
som_Latn	Somali	Latn	Afro-Asiatic
sot_Latn	Southern Sotho	Latn	Atlantic-Congo
spa_Latn	Spanish	Latn	Romance
srp_Cyrl	Serbian	Cyrl	Balto-Slavic
ssw_Latn	Swati	Latn	Atlantic-Congo
sun_Latn	Sundanese	Latn	Austronesian
swe_Latn	Swedish	Latn	Germanic
swh_Latn	Swahili	Latn	Atlantic-Congo
tam_Taml	Tamil	Taml	Dravidian
tel_Telu	Telugu	Telu	Dravidian
tgk_Cyrl	Tajik	Cyrl	Iranian
tgl_Latn	Tagalog	Latn	Austronesian
tha_Thai	Thai	Thai	Kra-Dai
tir_Ethi	Tigrinya	Ethi	Afro-Asiatic
tsn_Latn	Tswana	Latn	Atlantic-Congo
tso_Latn	Tsonga	Latn	Afro-Asiatic
tur_Latn	Turkish	Latn	Turkic
ukr_Cyrl	Ukrainian	Cyrl	Balto-Slavic
urd_Arab	Urdu	Arab	Indo-Aryan
urd_Latn	Urdu (Romanized)	Latn	Indo-Aryan
uzn_Latn	Northern Uzbek	Latn	Turkic
vie_Latn	Vietnamese	Latn	Austroasiatic
war_Latn	Waray	Latn	Austronesian
wol_Latn	Wolof	Latn	Atlantic-Congo
xho_Latn	Xhosa	Latn	Atlantic-Congo
yor_Latn	Yoruba	Latn	Atlantic-Congo
zho_Hans	Chinese (Simplified)	Hans	Sino-Tibetan
zho_Hant	Chinese (Traditional)	Hant	Sino-Tibetan
zsm_Latn	Standard Malay	Latn	Austronesian
zul_Latn	Zulu	Latn	Atlantic-Congo

Table 4: The 122 Languages & Scripts in BELEBELE.

As mentioned in Section 3, Bengali, Hindi, Sinhala, Nepali, Urdu, and Modern Standard Arabic are present twice, once in their respective native scripts and once in the Latin script. Chinese is also present twice, in Simplified and Traditional characters. There are 50 Indo-European languages, which we decide to display in smaller language families. Even so, Indo-Aryan is the most common language family (17), followed by Atlantic-Congo (16) and Afro-Asiatic (16).

Note that the language code used is from FLORES-200, and is not exactly the same as the older

FLORES-101 code, see the [FLORES website](#) for details.

A.2 Training Set

To create a training and development set to enable model finetuning to the task for adequate evaluation, we considered a diverse set of multiple-choice question-answering datasets. As there were little options for similarly-formatted datasets outside of English, we only selected English datasets.

After considering 33 different MRC datasets, we determine the most compatible to be RACE ([Lai et al., 2017](#)), SCIQ ([Welbl et al., 2017](#)), MULTIRC ([Khashabi et al., 2018](#)), MCTEST ([Richardson et al., 2013](#)), MCScript2.0 ([Ostermann et al., 2019](#)), and RECLOR ([Yu et al., 2020](#)). For each of the 6 datasets, we unpack and restructure the passages and questions from their respective formats. Some required more reformatting, such as MCScript2.0, which only has two multiple choice options. For this dataset, we sampled answers from different questions to complete 4 candidate answers. For each, we then filter out less suitable samples, such as questions with multiple correct answers (as in MultiRC), excessively long passages, or fill-in-the-blank questions. We then created subgroups of questions based of surface characteristics such as passage length, question length, and topic. We experiment with these different strata to train the best ROBERTA-base model ([Liu et al., 2019](#)) evaluated on the English set from BELEBELE in order to empirically validate the ability to teach a model the correct task. We use these empirical evaluations to finalize the training set, along with a development set of associated samples from the respective validation sets of the 6 above datasets. No test sets were included.

In the end, the dataset comprises 67.5k training samples and 3.7k development samples, more than half of which are from RACE. We provide a script at [our github repo](#) to reconstruct this dataset for anyone to perform task finetuning.

Note: Since the training set is a joint sample of other datasets, it is governed by a different license than BELEBELE. Most importantly, we do not claim any of that work or datasets to be our own. See Appendix A.3.

A.3 Licensing

The BELEBELE dataset is licensed under CC-BY-SA, as is the case for the underlying FLORES-200. Please refer to [our github repo](#) for more information.

The training set and assembly code is, however, licensed differently. The majority of the training set (data and code) is licensed under CC-BY-NC, therefore the use of BELEBELE for commercial purposes requires a different data solution for model finetuning.

A.4 Experiment Details

A.4.1 Model fine-tuning

As discussed in Section 4.2, for the fine-tuning of MLMs, we use the training set detailed in Appendix A.2. For all settings, the training was performed using the HuggingFace transformers library. We use the development set for hyperparameter search and evaluate the two best training runs on the BELEBELE splits.

For fine-tuning on the English training set, we find that 3 or 4 epochs is optimal for performance on the development set.

For Translate-Train-All, we use machine translation on passages, questions, and answers *separately*. Since there is almost 100x available data for this setting, we limit the training and validation sample to 650k and only train one epoch. We find that beyond this, there is not much improvement on the translated development set.

We provide fine-tuning hyperparameters of the reported runs below in Appendix A.7.1

A.4.2 In-Context Learning Prompt

As stated in Section 4.2, for 5-shot in-context learning, examples are sampled from the English training set and prompted to the model. The template used is as follows:

```
P: <passage> \n Q: <question> \n A: <mc answer 1> \n B: <mc answer 2> \n C: <mc answer 3> \n D: <mc answer 4> \n Answer: <Correct answer letter>
```

Within the answers {A, B, C, D}, we determine the prediction to be the one with the highest probability (relative to the others). For all our results, we report the average score over 3 runs.

A.4.3 Zero-Shot Instructions

As stated in Section 4.2, we evaluate both GPT3.5 and LLAMA-2-CHAT in the zero-shot setting by describing the task in natural language. We present the passage, question, and four possible answers, and instruct the model to provide the letter “A”, “B”, “C” or “D” as the answer. The instructions are given in English for all languages. We perform post-processing steps and accept answers predicted as e.g. “(A)” instead of “A”. The instructions and post-processing code are provided at [our github repo](#).

For the *In-language Translated Instructions* setting, we replicate the above, except present LLAMA-2-CHAT (70B) with instructions that are machine translated to the target language. We do not translate (or transliterate) the lettered answers we ask for (“A”, “B”, “C” or “D”). Therefore, the post-processing steps are the same.

For the *Translate-Test* setting, we present the natural language instructions in English. The passage, questions, and answers, however, have been machine-translated *individually* back to English.

A.5 Annotation Guidelines

A.5.1 MCQA Annotation Guidelines

The following is an abridged version of the particularized instructions provided to annotators for the task of creating a multiple-choice question-answering dataset. As mentioned, we additionally provided a positive and negative example for each guidelines to the annotators.

1. Ensure that all answers, if not most, are decently plausible to require the test-taker to fully read and understand the passage.
2. In order to make the questions not overly easy, ensure that if the correct answer is word-for-word from the passage, at least some of the wrong answers are as well. This is to ensure that the person answering can’t just guess the right answer based off identifying the answer in the passage through skimming.
3. Make the questions as specific as possible, leave nothing to ambiguity.
4. It should not be possible to answer the question without having read the passage, but the question must be answerable using just the passage (no external knowledge). We encourage the use of phrases such as “According to the passage...” in the questions if there may be ambiguity.
5. Try to sometimes have the answers be word-for-word in the passage and for other questions, the answers be in your own words. We would like a balance of these two categories of question.
6. Don’t write questions with double negatives that may trivially fool the person answering.
7. Try to ask questions in your own words and don’t copy and paste entire phrases from the paragraph. This allows us to truly evaluate the comprehension as opposed to recognizing patterns in the way the question is extracted from the text. That being said, making the questions comprehensible and including details is very important, as mentioned above.
8. Avoid using ambiguous qualifiers (for example, “very” or “really”) or having the question centered around an extreme (for example, “the greatest” or “the most”). These phrases may leave room for

ambiguity and subjectivity. In addition, qualified statements change with time and may not be valid in the future.

A.5.2 Translation Specifications

To align the dialect or language variant to FLORES, we instructed the LSP to use the same localization as in the FLORES-200 creation a few years prior. To align style, formality, and wording, we supplemented the traditional translation guidelines with the following:

Given that the associated translated passage is already established (and not subject to change as it is in a published dataset), the translations of the questions and answers have to be fully compatible with it. This means that for dates, proper nouns, units of measure, etc. where there is potential ambiguity, the translators have to follow what was done for the passage, even if they disagree that it is the more correct translation.

For example,

Hungarian Translated passage: “Jesus készen áll a vitára. . .”

English question: “What was Jesus working on when. . .”

Therefore, in the questions and answers Jesus must be translated as “Jesus” and not “Jézus”.

A.6 MCQA Lexical Featurization

As discussed in Section 3, we use lexical featurization to programmatically assess the ease of solvability of our multiple-choice questions. These features allow us to then run statistical tests and determine if there are implicit patterns (or “bias”) that models can learn and exploit.

In the list below, note that lexical overlap was measured using 1-, 2-, 3-, and 4-grams.

1. The frequency of lexical overlap between the passage and the correct answer, in comparison to overlap with wrong answers. This is to ensure the question cannot be answered by selecting the only options present in the original passage.
2. The frequency of lexical overlap between the question and the correct answer, in comparison to overlap with wrong answers. This is to ensure the question cannot be answered by selecting options nearly extract-able from the question itself.
3. If the question has lexical overlap with a sentence in the passage, is the correct answer in that same sentence, in comparison to overlap with wrong answers? This is to ensure the question cannot be answered by finding the sentence which the question is alluding to and extracting the answer from there directly.
4. The frequency of a correct answer to have lexical overlap with the question, in comparison to wrong answers. Overlap with the question may giveaway the answer.
5. The frequency of a correct answer to have lexical overlap with other answer options, in comparison to a pair of wrong answers. If the correct answer is more likely to have closer neighbors, it could allow models to narrow down options quicker.
6. Simple length statistics (character and word count) of correct answer, in comparison to wrong answers

A.7 Detailed Results Tables

A.7.1 Cross-Lingual MLMs

Full Results for Cross-Lingual MLMs						
Evaluation Model Name Size/Variant	Finetune in English			Translate-Train-All		
	XLM-V large	INFOXML large	XLM-R large	XLM-V large	INFOXML large	XLM-R large
AVG	55.6	56.2	54.0	60.2	60.0	58.9
PCT Above 50	69.7%	67.2%	64.8%	76.2%	70.5%	69.7%
PCT Above 70	21.9%	28.9%	15.7%	33.1%	37.2%	36.4%
eng_Latn	76.2	79.3	76.2	77.8	81.2	78.7
acm_Arab	51.2	57.3	55.4	55.3	57.6	59.2
afr_Latn	69.3	72.7	69.1	72.3	75.1	74.3
als_Latn	68.4	68.9	64.9	70.8	72.2	71.4
amh_Ethi	53.1	52.9	52.6	61.6	60.0	60.7
apc_Arab	56.1	58.8	57.9	57.7	60.6	61.9
arb_Arab	67.2	71.0	69.8	70.6	75.0	74.3
arb_Latn	29.3	32.2	27.6	31.6	33.4	30.6
ars_Arab	55.6	59.9	58.9	61.1	65.8	65.9
ary_Arab	43.8	48.7	44.0	48.0	52.8	52.6
arz_Arab	56.9	60.2	57.6	61.4	64.9	66.1
asm_Beng	53.7	53.6	49.3	58.6	58.8	56.9
azj_Latn	59.7	61.3	59.0	65.0	65.6	65.1
bam_Latn	34.2	34.9	33.2	39.2	39.1	36.9
ben_Beng	60.0	63.4	59.6	65.6	69.6	63.7
ben_Latn	46.8	36.9	38.8	53.0	42.7	48.1
bod_Tibt	24.0	24.9	23.7	24.8	23.3	36.9
bul_Cyrl	72.6	72.0	70.1	74.0	75.3	74.2
cat_Latn	71.6	74.4	72.0	75.7	78.1	74.7
ceb_Latn	45.4	44.1	42.3	52.0	52.6	50.7
ces_Latn	69.9	72.3	69.9	72.3	76.2	74.4
ckb_Arab	29.7	52.3	30.3	36.9	58.0	36.9
dan_Latn	70.8	74.1	72.9	73.0	76.3	74.7
deu_Latn	72.6	75.7	72.9	74.1	78.7	76.7
ell_Grek	70.3	72.3	70.3	73.1	74.9	73.0
est_Latn	63.2	67.2	64.8	68.7	70.7	70.4
eus_Latn	63.6	66.1	64.8	68.2	70.8	70.3
fin_Latn	69.1	72.4	72.2	73.0	75.2	74.9
fra_Latn	73.1	74.2	72.1	74.6	76.8	75.6
fuv_Latn	29.7	27.7	26.4	32.8	30.7	31.1
gaz_Latn	48.8	33.8	36.4	52.6	36.0	43.3
grn_Latn	53.9	37.8	37.9	59.6	40.6	41.9
guj_Gujr	58.7	57.0	54.1	63.3	65.9	63.1
hat_Latn	57.1	39.6	35.2	63.2	44.1	39.8
hau_Latn	51.0	41.1	48.2	53.4	48.1	53.0
heb_Hebr	67.2	68.2	64.8	69.3	72.3	70.6
hin_Deva	57.9	60.2	57.4	63.8	64.3	63.4
hin_Latn	53.1	49.7	46.8	57.6	55.4	58.9
hrv_Latn	70.0	72.4	69.9	71.2	75.3	74.0
hun_Latn	69.7	70.8	70.0	73.1	74.2	72.8
hye_Armn	59.4	61.0	58.9	65.9	66.1	64.7
ibo_Latn	40.1	32.2	31.2	46.8	32.0	32.2
ilo_Latn	37.4	36.3	33.8	38.1	40.6	39.7
ind_Latn	68.9	70.7	68.0	71.3	73.1	70.4
isl_Latn	67.3	66.0	63.8	70.1	68.9	69.0
ita_Latn	70.6	72.8	70.0	71.8	76.4	73.3
jav_Latn	64.2	59.8	60.8	67.2	63.3	66.8
jpn_Jpan	66.4	70.1	67.6	71.3	71.8	71.0
kac_Latn	32.0	29.1	32.1	33.8	34.0	33.3
kan_Knda	61.1	62.0	59.7	66.6	68.4	69.1
kat_Geor	64.7	64.8	63.6	68.0	68.9	67.4
kaz_Cyrl	60.1	61.6	56.8	64.9	65.3	64.7
kea_Latn	44.0	45.2	44.9	48.7	47.7	48.1
khk_Cyrl	56.7	58.8	57.8	61.1	64.6	64.2
khm_Khmr	60.0	59.0	57.7	63.0	64.2	63.8
kin_Latn	35.9	33.6	34.3	39.1	39.1	38.6
kir_Cyrl	65.4	63.4	61.8	68.3	68.2	67.7
kor_Hang	70.1	71.4	68.7	72.9	74.6	74.8
lao_Lao	55.8	57.6	53.0	63.2	63.6	63.0
lin_Latn	44.7	33.2	30.6	50.9	35.3	34.4

lit_Latn	68.3	69.4	67.2	71.7	72.9	72.0
lug_Latn	39.9	29.4	31.6	47.8	34.7	34.7
luo_Latn	30.3	30.9	30.8	33.7	34.9	33.2
lvs_Latn	70.1	71.3	68.7	74.1	75.6	73.0
mal_Mlym	62.0	65.0	62.7	69.1	68.3	67.1
mar_Deva	62.6	65.2	60.8	69.2	68.8	67.2
mkd_Cyrl	67.8	69.3	65.7	71.0	73.8	72.8
mlt_Latn	37.9	57.1	38.1	40.2	63.7	42.7
mri_Latn	32.0	30.6	32.2	33.0	35.7	34.0
mya_Mymr	56.6	59.1	53.6	62.2	65.1	62.9
nld_Latn	68.4	71.7	71.0	68.6	74.0	72.8
nob_Latn	71.8	73.6	70.7	72.8	75.4	74.2
npi_Deva	58.4	60.7	55.7	64.4	65.8	62.7
npi_Latn	38.3	35.8	33.8	37.4	36.4	34.8
nso_Latn	45.9	31.3	30.0	53.2	34.1	34.7
nya_Latn	31.0	29.2	29.8	34.2	33.0	30.8
ory_Orya	60.8	62.1	58.6	65.6	65.4	63.9
pan_Guru	58.1	59.2	57.8	63.1	62.6	62.0
pbt_Arab	55.4	56.0	51.0	60.6	62.6	61.1
pes_Arab	68.3	69.1	68.2	70.8	73.6	72.0
plt_Latn	55.7	45.6	52.7	61.7	53.4	58.1
pol_Latn	69.0	70.4	67.4	72.1	73.7	72.7
por_Latn	70.9	74.3	70.6	73.8	77.1	74.0
ron_Latn	72.3	72.9	71.3	74.0	76.2	74.8
rus_Cyrl	71.9	73.8	72.2	75.4	76.8	77.1
shn_Mymr	26.9	25.2	26.3	25.0	26.4	27.0
sin_Latn	24.9	34.2	30.7	41.7	38.3	37.3
sin_Sinh	64.4	67.2	62.7	69.8	70.2	68.6
slk_Latn	69.3	71.9	70.2	72.6	76.7	73.0
slv_Latn	69.7	72.2	68.6	71.8	75.4	73.9
sna_Latn	34.8	37.2	33.2	37.1	38.6	35.9
snd_Arab	55.2	56.6	51.9	60.0	61.3	61.3
som_Latn	46.0	39.1	42.6	50.7	46.3	50.7
sot_Latn	46.8	29.3	31.3	52.0	31.9	32.7
spa_Latn	71.0	73.3	71.4	72.7	75.3	76.4
srp_Cyrl	71.0	70.9	71.1	73.6	76.1	75.9
ssw_Latn	39.8	30.6	34.3	47.1	34.3	38.9
sun_Latn	60.9	50.7	55.3	64.2	55.8	59.4
swe_Latn	73.0	75.0	74.2	74.2	76.9	75.1
swl_Latn	64.9	65.3	62.8	69.3	69.2	68.7
tam_Taml	61.8	64.6	61.7	67.4	69.4	65.3
tel_Telu	55.6	57.8	53.6	62.1	63.2	61.1
tgk_Cyrl	38.2	58.6	33.8	39.2	64.3	39.6
tgl_Latn	69.2	67.4	64.7	72.0	70.4	70.0
tha_Thai	63.8	68.1	65.8	69.0	68.9	70.1
tir_Ethi	33.3	36.7	33.8	39.9	42.1	37.7
tsn_Latn	49.0	35.0	30.8	49.8	35.7	34.3
tso_Latn	37.9	36.3	34.2	41.7	39.7	37.1
tur_Latn	66.7	70.2	66.8	70.6	72.0	72.0
ukr_Cyrl	70.4	70.9	71.0	72.3	74.9	75.0
urd_Arab	61.6	63.8	59.3	65.6	68.6	66.3
urd_Latn	42.2	42.6	40.8	49.4	48.9	48.4
uzn_Latn	65.2	66.9	64.4	69.1	70.6	70.2
vie_Latn	69.6	71.1	69.4	73.7	72.9	71.4
war_Latn	46.4	44.7	43.7	47.6	49.3	46.6
wol_Latn	36.8	32.2	30.4	40.6	32.3	32.2
xho_Latn	48.7	36.1	39.0	54.4	40.2	45.4
yor_Latn	35.0	29.3	28.7	38.6	32.0	27.9
zho_Hans	69.8	74.6	71.0	73.7	76.2	74.8
zho_Hant	69.2	72.4	67.1	73.1	74.3	71.3
zsm_Latn	69.1	72.6	69.9	72.4	73.3	72.2
zul_Latn	46.9	36.4	39.0	54.2	39.8	44.1

Table 5: Results of Cross-Lingual MLMs in the two settings described in Section 4.

For all, the *large* version was used which is the same architecture across all three. XLM-V has a significantly larger vocabulary size, leading to more total parameters. We find that in general, INFOXLM and XLM-V are very similar and both out-perform XLM-R across the board. INFOXLM outperforms

XLM-V in higher- and medium-resource languages, while XLM-V performs better on the lowest-resource languages. As a result, XLM-V has the most scores above 50, but INFOXLM has more scores above 70. In the below table, we provide the hyperparameters used for these specific runs.

Full Results for Cross-Lingual MLMs						
Evaluation Model Name Size/Variant	Finetune in English			Translate-Train-All		
	XLM-V large	INFOXLM large	XLM-R large	XLM-V large	INFOXLM large	XLM-R large
epochs	3	4	3	1	1	1
training set size	67.5k	67.5k	67.5k	650k	650k	650k
learning rate	5e-6	4e-6	5e-6	3-6	3e-6	3e-6
weight decay	0.01	0.01	0.01	0.001	0.001	0.001
batch size	64	64	64	64	64	64

Table 6: Finetuning Hyperparameters for the runs reported above. More details in Appendix A.4.

A.7.2 LLMs

Full Results for Large Language Models						
Evaluation Model Name Size/Variant	Zero-Shot for Instructed Models		5-shot In-Context Learning			Translate-Train-All XLM-V large
	GPT3.5-TURBO	LLAMA-2-CHAT 70B	LLAMA 2 70B	LLAMA 1 65B	FALCON 40B	
AVG	50.6	41.5	48.0	40.9	37.3	60.2
PCT Above 50	43.4%	27.1%	38.5%	25.4%	16.4%	76.2%
PCT Above 70	28.9%	2.5%	26.2%	12.3%	1.6%	33.1%
eng_Latn	87.7	78.8	90.9	82.5	77.2	77.8
acm_Arab	51.6	35.9	47.9	37.9	37.6	55.3
afr_Latn	78.3	57.9	75.9	60.7	53.4	72.3
als_Latn	67.1	36.0	45.4	34.9	36.6	70.8
amh_Ethi	28.7	28.9	27.5	27.8	24.8	61.6
apc_Arab	55.6	38.8	51.2	39.6	36.3	57.7
arb_Arab	69.3	42.3	61.7	44.1	38.3	70.6
arb_Latn	31.1	30.2	26.8	28.0	26.3	31.6
ars_Arab	55.1	37.4	50.2	40.7	32.1	61.1
ary_Arab	45.7	32.6	40.6	33.1	32.3	48.0
arz_Arab	56.7	37.3	50.7	37.4	33.0	61.4
asm_Beng	36.0	35.7	32.3	28.9	22.4	58.6
azj_Latn	54.9	33.4	42.2	33.6	34.1	65.0
bam_Latn	31.7	29.4	30.3	28.4	29.7	39.2
ben_Beng	43.6	34.9	39.1	33.4	22.6	65.6
ben_Latn	34.6	30.4	29.6	29.2	32.1	53.0
bod_Tibt	26.6	28.3	25.7	24.9	26.8	24.8
bul_Cyrl	76.0	65.0	80.4	69.3	41.9	74.0
cat_Latn	78.4	68.2	84.6	76.3	58.8	75.7
ceb_Latn	53.3	40.6	50.4	38.9	39.2	52.0
ces_Latn	76.9	65.0	81.1	70.7	65.0	72.3
ckb_Arab	31.8	32.8	28.7	31.6	28.9	36.9
dan_Latn	80.7	66.2	83.6	73.6	56.2	73.0
deu_Latn	83.3	69.4	84.6	76.0	70.1	74.1
ell_Grek	73.0	50.7	64.9	44.2	31.2	73.1
est_Latn	73.1	36.6	53.0	36.3	34.9	68.7
eus_Latn	40.9	31.1	34.7	32.8	38.9	68.2
fin_Latn	77.9	62.7	79.3	55.7	42.8	73.0
fra_Latn	83.1	72.2	86.4	77.5	69.7	74.6
fuv_Latn	26.1	29.8	24.9	25.4	25.1	32.8
gaz_Latn	30.3	29.3	27.8	29.1	24.9	52.6
grn_Latn	34.2	32.2	32.4	30.3	33.8	59.6
guj_Gujr	38.4	31.1	27.1	25.7	24.7	63.3
hat_Latn	51.6	34.1	37.4	33.7	36.2	63.2
hau_Latn	32.2	32.1	28.0	26.4	28.9	53.4
heb_Hebr	64.2	41.4	54.9	41.4	31.1	69.3
hin_Deva	49.1	42.0	52.6	38.4	27.1	63.8
hin_Latn	52.3	39.2	49.0	34.2	40.0	57.6

hrv_Latn	78.4	64.7	79.8	66.9	48.7	71.2
hun_Latn	74.6	61.1	78.8	66.7	37.7	73.1
hye_Armn	35.0	31.9	34.1	32.1	25.4	65.9
ibo_Latn	28.4	30.1	27.4	25.3	30.2	46.8
ilo_Latn	37.1	33.2	36.6	32.1	35.1	38.1
ind_Latn	74.2	61.3	81.4	55.7	52.1	71.3
isl_Latn	62.3	38.0	54.3	42.1	36.4	70.1
ita_Latn	80.0	68.6	84.5	76.1	66.4	71.8
jav_Latn	46.7	37.0	40.3	33.0	36.8	67.2
jpn_Jpan	70.9	56.6	77.6	53.9	49.6	71.3
kac_Latn	30.9	30.7	27.7	28.6	27.8	33.8
kan_Knda	40.6	21.9	25.7	24.4	24.0	66.6
kat_Geor	33.0	34.6	37.8	34.3	23.4	68.0
kaz_Cyrl	35.0	32.4	29.3	32.4	32.6	64.9
kea_Latn	46.0	38.1	45.4	38.1	38.0	48.7
khk_Cyrl	32.0	31.1	29.8	28.4	27.4	61.1
khm_Khmr	30.4	30.6	27.0	28.2	25.0	63.0
kin_Latn	35.2	30.6	29.8	28.5	31.9	39.1
kir_Cyrl	37.9	32.2	34.6	32.5	31.9	68.3
kor_Hang	67.1	56.3	77.8	52.9	40.2	72.9
lao_Lao	30.0	26.5	24.3	26.2	28.1	63.2
lin_Latn	33.8	31.0	28.0	30.4	29.3	50.9
lit_Latn	72.0	39.7	52.1	39.6	39.3	71.7
lug_Latn	28.4	30.9	29.2	28.3	28.9	47.8
luo_Latn	27.1	31.2	29.4	29.3	29.9	33.7
lvs_Latn	70.8	41.0	51.3	39.0	37.6	74.1
mal_Mlym	34.9	30.1	32.4	30.0	21.2	69.1
mar_Deva	38.3	34.8	41.2	32.9	25.0	69.2
mkd_Cyrl	69.4	55.7	72.5	56.2	38.1	71.0
mlt_Latn	44.8	36.2	44.9	36.7	35.4	40.2
mri_Latn	33.3	31.8	28.5	32.0	29.7	33.0
mya_Mymr	30.3	31.3	24.1	24.2	22.6	62.2
nld_Latn	80.4	66.2	82.2	73.3	66.7	68.6
nob_Latn	79.0	65.7	81.8	70.9	60.8	72.8
npi_Deva	40.4	32.9	40.4	33.0	25.4	64.4
npi_Latn	35.1	30.4	30.2	30.0	30.9	37.4
nso_Latn	33.6	30.1	30.4	27.4	29.3	53.2
nya_Latn	33.2	29.3	27.3	28.7	29.3	34.2
ory_Orya		29.2	24.8	23.9	23.7	65.6
pan_Guru	39.1	33.1	26.3	27.1	23.4	63.1
pbt_Arab	32.3	30.2	30.8	29.4	29.4	60.6
pes_Arab	61.8	41.8	53.9	41.0	35.9	70.8
plt_Latn	32.3	30.5	29.6	31.0	31.4	61.7
pol_Latn	74.7	61.7	79.2	67.0	59.9	72.1
por_Latn	83.0	70.2	86.1	75.4	68.3	73.8
ron_Latn	77.4	65.6	83.4	73.2	66.6	74.0
rus_Cyrl	78.4	67.0	82.7	73.1	48.1	75.4
shn_Mymr		28.2	25.6	22.7	24.0	25.0
sin_Latn	30.4	31.9	33.8	27.9	32.6	41.7
sin_Sinh	32.6	33.4	25.2	29.4	27.7	69.8
slk_Latn	77.3	58.8	75.2	60.4	57.0	72.6
slv_Latn	77.4	62.4	76.7	65.6	43.7	71.8
sna_Latn	35.4	30.2	27.4	28.3	31.6	37.1
snd_Arab	34.1	29.7	30.9	28.9	30.2	60.0
som_Latn	32.4	30.3	27.8	27.6	29.9	50.7
sot_Latn	33.9	30.0	28.9	26.8	29.9	52.0
spa_Latn	79.2	68.4	85.0	74.8	69.2	72.7
srp_Cyrl	74.8	65.1	81.0	70.7	40.2	73.6
ssw_Latn	32.0	30.7	27.7	28.0	30.1	47.1
sun_Latn	38.9	34.9	37.8	30.7	34.1	64.2
swe_Latn	81.7	67.4	82.7	73.7	67.3	74.2
swh_Latn	70.3	35.1	39.6	34.4	36.7	69.3
tam_Taml	32.8	34.4	33.2	31.6	24.4	67.4
tel_Telu	34.6	27.5	25.9	26.6	22.4	62.1
tgk_Cyrl	37.7	32.5	34.0	33.1	32.7	39.2
tgl_Latn	66.7	49.6	68.1	48.3	47.7	72.0
tha_Thai	55.7	38.9	46.2	35.0	33.0	69.0
tir_Ethi	28.4	29.6	24.5	23.5	25.0	39.9
tsn_Latn	31.8	30.1	28.5	24.7	31.2	49.8
tso_Latn	33.4	30.0	30.4	28.0	29.7	41.7

tur_Latn	69.9	47.3	65.4	42.1	39.6	70.6
ukr_Cyrl	72.8	65.7	80.8	69.7	41.9	72.3
urd_Arab	48.3	37.0	43.2	34.7	31.7	65.6
urd_Latn	40.3	34.1	38.0	30.1	34.2	49.4
uzn_Latn	44.1	33.1	35.1	30.6	33.1	69.1
vie_Latn	72.9	59.6	78.4	43.5	41.4	73.7
war_Latn	48.9	39.3	44.4	37.4	38.6	47.6
wol_Latn	29.0	28.9	27.6	26.0	26.8	40.6
xho_Latn	30.0	29.9	28.2	27.6	30.2	54.4
yor_Latn	29.1	30.1	28.3	27.7	27.2	38.6
zho_Hans	77.6	62.4	83.7	64.6	66.0	73.7
zho_Hant	76.3	59.3	82.0	57.7	62.2	73.1
zsm_Latn	74.0	56.4	76.3	51.7	51.3	72.4
zul_Latn	30.4	30.2	29.7	27.1	30.7	54.2

Table 7: Results on LLMs, with comparison to full finetuning on XLM-V

The evaluation settings and models are described in more detail in Section 4. We see that none of these models can understand many of the 122 languages, while demonstrating excellent performance on high-resource languages. The 175B-parameter GPT3.5-TURBO outperforms LLAMA-2-CHAT (70B) across the board and has comparable results to LLAMA 2 (70B) even though it is in zero-shot. Note that GPT3.5-TURBO threw errors when processing characters in Shan (shn_Mymr) and Oriya (ory_Orya) and therefore we could not evaluate the results. For the purposes of aggregated scores, we consider this a score of 25.0. For comparison to fully-finetuned multilingual models, we re-provide the results of XLM-V-large.

A.7.3 Languages in Multiple Scripts

Comparative Results for Languages with Multiple Scripts					
Evaluation Model Name	Zero-Shot	Five-Shot		Finetune in English	AVG
	GPT3.5-TURBO	LLAMA 2 (70B)	FALCON (40B)	INFOXML-large	
arb_Arab	69.3	61.7	38.3	71.0	60.1
arb_Latn	31.1	26.8	26.3	32.2	29.1
ben_Beng	43.6	39.1	22.6	63.4	42.2
ben_Latn	34.6	29.6	32.1	36.9	33.3
hin_Deva	49.1	52.6	27.1	60.2	47.3
hin_Latn	52.3	49.0	40.0	49.7	47.8
npi_Deva	40.4	40.4	25.4	60.7	41.7
npi_Latn	35.1	30.2	30.9	35.8	33.0
sin_Sinh	32.6	25.2	27.7	67.2	38.2
sin_Latn	30.4	33.8	32.6	34.2	32.8
urd_Arab	48.3	43.2	31.7	63.8	46.7
urd_Latn	40.3	38.0	34.2	42.6	38.8
zho_Hant	76.3	82.0	62.2	72.4	73.3
zho_Hans	77.6	83.7	66.0	74.6	75.5

Table 8: Selected results from 3 models in differing settings comparing languages present in multiple scripts.

We find that generally, the performance is higher in the native script than the romanized version, except for FALCON which displays the opposite trend on the 5 Indo-Aryan languages. In Chinese, performance on simplified & traditional are very similar with simplified being higher across all 4. For INFOXML, we display the English finetuning score.

A.7.4 Translate-Test

Translate-Test Results on 91 languages
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Evaluation Model	Zero-Shot LLAMA-2-CHAT (70B)	Translate-Test, Zero-Shot	Translate-Train-All XLM-V-large
AVG	44.0	57.1	64.9
PCT Above 50	35.2%	78.0%	90.1%
PCT Above 70	2.2%	2.2%	42.9%
eng_Latn	78.8	78.8	77.8
fra_Latn	72.2	70.6	73.1
por_Latn	70.2	69.9	70.9
deu_Latn	69.4	65.7	72.6
ita_Latn	68.6	66.1	70.6
spa_Latn	68.4	69.3	71.0
cat_Latn	68.2	67.0	71.6
swe_Latn	67.4	66.1	73.0
rus_Cyrl	67.0	67.3	71.9
dan_Latn	66.2	66.8	70.8
nld_Latn	66.2	67.2	68.4
nob_Latn	65.7	68.3	71.8
ukr_Cyrl	65.7	66.0	70.4
ron_Latn	65.6	67.0	72.3
srp_Cyrl	65.1	66.2	71.0
bul_Cyrl	65.0	67.7	72.6
ces_Latn	65.0	65.6	69.9
hrv_Latn	64.7	65.3	70.0
fin_Latn	62.7	61.1	69.1
slv_Latn	62.4	61.2	69.7
zho_Hans	62.4	71.2	69.8
pol_Latn	61.7	63.0	69.0
ind_Latn	61.3	64.8	68.9
hun_Latn	61.1	62.9	69.7
vie_Latn	59.6	59.4	69.6
zho_Hant	59.3	65.8	69.2
slk_Latn	58.8	66.2	69.3
afr_Latn	57.9	65.0	69.3
jpn_Jpan	56.6	54.8	66.4
zsm_Latn	56.4	67.0	69.1
kor_Hang	56.3	56.7	70.1
mkd_Cyrl	55.7	66.7	67.8
ell_Grek	50.7	67.6	70.3
tgl_Latn	49.6	62.2	69.2
tur_Latn	47.3	62.6	66.7
arb_Arab	42.3	60.7	67.2
hin_Deva	42.0	62.8	57.9
pes_Arab	41.8	59.6	68.3
heb_Hebr	41.4	62.0	67.2
lvs_Latn	41.0	60.9	70.1
ceb_Latn	40.6	62.6	45.4
lit_Latn	39.7	60.8	68.3
hin_Latn	39.2	52.7	53.1
tha_Thai	38.9	54.1	63.8
isl_Latn	38.0	58.1	67.3
jav_Latn	37.0	55.3	64.2
urd_Arab	37.0	59.4	61.6
est_Latn	36.6	59.4	63.2
als_Latn	36.0	63.1	68.4
asm_Beng	35.7	57.7	53.7
swh_Latn	35.1	57.8	64.9
ben_Beng	34.9	61.0	60.0
sun_Latn	34.9	50.8	60.9
mar_Deva	34.8	60.0	62.6
kat_Geor	34.6	57.7	64.7
tam_Taml	34.4	55.9	61.8
urd_Latn	34.1	43.0	42.2
hat_Latn	34.1	56.3	57.1
azj_Latn	33.4	55.6	59.7
sin_Sinh	33.4	57.7	64.4
pan_Guru	33.1	57.6	58.1
npi_Deva	32.9	62.0	58.4
ckb_Arab	32.8	51.3	29.7
kaz_Cyrl	32.4	53.2	60.1

hau_Latn	32.1	43.4	51.0
hye_Armn	31.9	58.0	59.4
mya_Mymr	31.3	46.6	56.6
khk_Cyrl	31.1	52.2	56.7
guj_Gujr	31.1	59.6	58.7
lin_Latn	31.0	40.3	44.7
lug_Latn	30.9	38.7	39.9
ssw_Latn	30.7	43.2	39.8
khm_Khmr	30.6	52.8	60.0
plt_Latn	30.5	46.7	55.7
ben_Latn	30.4	45.1	46.8
som_Latn	30.3	40.8	46.0
pbt_Arab	30.2	48.8	55.4
zul_Latn	30.2	44.4	46.9
nso_Latn	30.1	43.4	45.9
tsn_Latn	30.1	40.4	49.0
yor_Latn	30.1	37.7	35.0
ibo_Latn	30.1	35.3	40.1
mal_Mlym	30.1	63.0	62.0
xho_Latn	29.9	49.2	48.7
fuv_Latn	29.8	29.4	29.7
gaz_Latn	29.3	37.0	48.8
ory_Orya	29.2	57.8	60.8
amh_Ethi	28.9	50.4	53.1
wol_Latn	28.9	39.0	36.8
tel_Telu	27.5	54.3	55.6
lao_Lao	26.5	47.4	55.8
kan_Knda	21.9	62.0	61.1

Table 9: Comparing LLAMA-2-CHAT Zero-Shot performance In-Language vs Translate-Test on 91 languages, with an additional comparison to Translate-Train-All finetuning on XLM-V.

We evaluate 91 of the 122 languages in Translate-Test and find that Translate-Test performance dominates in-language performance on a big majority of languages for LLAMA-2-CHAT (70B) in zero-shot. A few head languages such as German and Italian have higher scores in the traditional setting, but nearly all medium-resource languages are better understood with machine translation. For nearly all low-resource languages, the difference is over 20 accuracy points. For comparison, we see that machine translation lifts LLAMA-2-CHAT performance in zero-shot not far from fully finetuned XLM-V-large. This is illustrated more clearly in (Fig. 5) below.

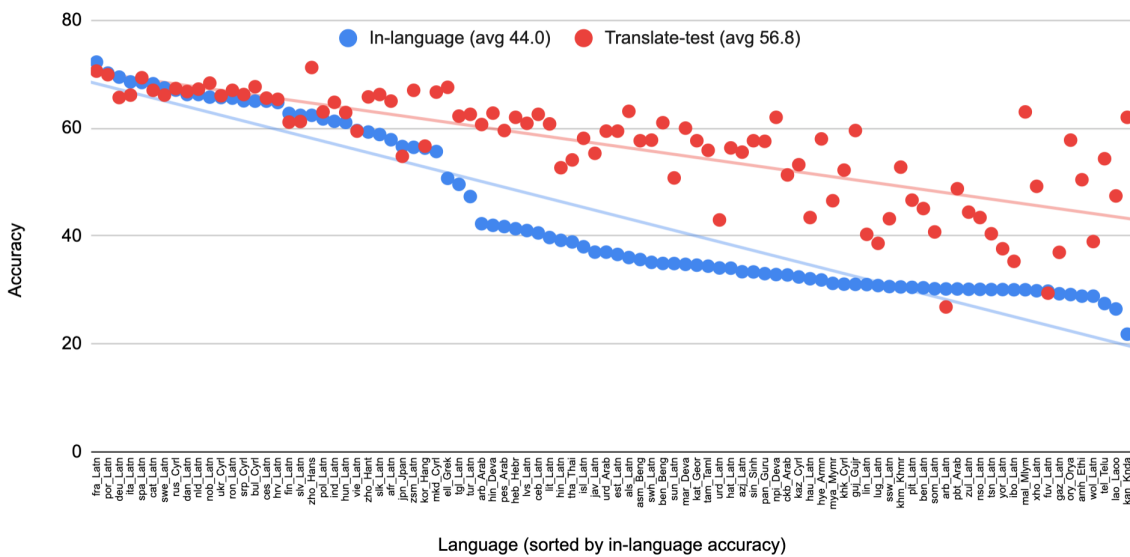


Figure 5: Comparison of LLAMA-2-CHAT (70B) zero-shot performance on Translate-Test and the standard in-language evaluation.